

JEE MAIN REVISION MASTER: STATES OF MATTER 2026 0 QUESTIONS 0% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 8 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Gaseous State — Core Principles & Governing Laws	6 PYQs	Core Principle
Standard	Gaseous State — Standard Calculations & Problem Archetypes	2 PYQs	Quantitative Analysis
Standard	Gaseous State — Applied Cases, Reactions & Traps	1 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Gaseous State — Core Principles & Governing Laws	$pV=RT$, HCO_3^- , $R=0.08210\text{LatmKmol}^{-1}$
Gaseous State — Standard Calculations & Problem Archetypes	$Z=\frac{V_{\text{real}}}{V_{\text{ideal}}}$
Gaseous State — Applied Cases, Reactions & Traps	HCO_3^- , CO_2

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Gaseous State — Core Principles & Governing Laws (Moderate Yield — 6 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Gaseous State.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Gaseous State.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Gaseous State PYQs.

Representative Source PYQs:

- [2022 Shift] Q#2 (MCQ): "Which property of CO_2 makes it biologically and geo-chemically important?..." [Ans: N/A]
- [2022 Shift] Q#3 (MCQ): "The volume of 2.8g of CO at 27°C and 0.821 atm pressure is ($R=0.08210\text{L. atm Kmol}^{-1}$)..." [Ans: N/A]
- [2021 Shift] Q#4 (MCQ): "When the absolute temperature of ideal gas is doubled and pressure is halved, the volume of gas..." [Ans: N/A]

3.2 Gaseous State — Standard Calculations & Problem Archetypes (Standard Yield — 2 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Gaseous State.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Identify the wrong relation for real gases :..." [Ans: N/A]
- [2017 Shift] Q#4 (MCQ): "The pressure of real gases is less than that of ideal gas because of..." [Ans: N/A]

3.3 Gaseous State — Applied Cases, Reactions & Traps (Standard Yield — 1 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Gaseous State.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2022 Shift] Q#2 (MCQ): "Which property of CO_2 makes it biologically and geo-chemically important?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2022 (Q#2)

Which property of CO_2 makes it biologically and geo-chemically important?

- (A) Its colourless and odourless nature
- (B) Its low solubility in water
- (C) Its high compressibility
- (D) Its acidic nature

Correct Answer: None

Solution Analysis: The property of low solubility of carbon dioxide in water makes it biologically and geo-chemically important. Carbon dioxide on reaction with water form carbonic acid, which dissociates to give HCO_3^- ions. $\text{H}_2\text{CO}_3/\text{HCO}_3^-$ buffer system helps to maintain pH of blood between 7.26-7.42 .

Benchmark Problem 2: KCET 2017 (Q#4)

The pressure of real gases is less than that of ideal gas because of

- (A) increases in the kinetic energy of the molecules
- (B) increases in the number of collisions
- (C) intermolecular attraction
- (D) finite size of particles

Correct Answer: None

Solution Analysis: The pressure is much greater in ideal gas compared to the pressure of a real gas since the particles do not have the attractive forces that enable the molecules to hold back when they will collide at an impact. Hence, particles collide with less energy.